

Software Validation and Benchmark Report

1. Automated Tests

Result: 5 passed, 0 failed (local test suite).
Components: fusion, error metrics, optimization, bootstrap, SVM.

2. Benchmark Configuration

Data: generated score pairs, NOT original HMOG recordings.
Three seeds: 7, 21, 42; 400 attempts for each seed.
Two modalities; 200 fusion-learning and 200 held-out attempts.
Optimizer: swarm size 20, 15 iterations; CFA levels 0.5, 1.0, 1.5.
Comparator: fixed unweighted sum, 0.5 acceptance threshold.
This comparator is NOT the published methods [24] or [25].

3. Mean Held-out Global Error

CFA	Optimized fusion	Fixed-sum baseline	Difference
0.5	0.2350	0.2142	+0.0208
1.0	0.2384	0.2194	+0.0190
1.5	0.2132	0.2246	-0.0114

Lower weighted global error is better. Optimization reduced cost for CFA = 1.5 but not for CFA = 0.5 or 1.0 in these generated-data runs.

4. Interpretation

Software tests passed; benchmark results vary with the security coefficient.
The example outputs do not establish published HMOG performance.

Run: python -m pytest -q
Run: python experiments/benchmark.py
Full numeric outputs: reports/benchmark_results.json

RUN-LEVEL BENCHMARK OUTPUTS

Seed	CFA	Rule	Threshold	Optimized E	Fixed E
7	0.5	product	0.4328	0.1710	0.1609
7	1.0	product	0.4622	0.2047	0.1597
7	1.5	exp	1.7678	0.2216	0.1585
21	0.5	exp	1.6918	0.2861	0.2644
21	1.0	exp	1.7177	0.2628	0.2788
21	1.5	hamacher	0.3680	0.2404	0.2933
42	0.5	exp	1.5925	0.2480	0.2174
42	1.0	sum	0.4796	0.2477	0.2197
42	1.5	sum	0.5797	0.1776	0.2220

5. SVM Checkpoints

Included models: touch_svm.json (3 features), motion_svm.json (4 features).

Model type: linear SVC with calibrated probability estimates.

Training source: generated example feature data; NOT HMOG.

6. Reading the Results

APSO and fixed-sum values are evaluated on the same synthetic holdout.

These results are descriptive and do not test statistical significance.

For manuscript replication, execute on real study features and partitions.

Related files:

experiments/benchmark.py models/ reports/benchmark_results.json